LECTURE PLAN

Institute/Department	UNIVERSITY INSTITUTE OF COMPUTING (UIC)	**Program**	**Master of Computer Applications - Artificial Intelligence & Machine Learning(MC306)**
Master Subject Coordinator Name:	Sameera Dhuria	**Master Subject Coordinator E-Code:**	**E17674**
Course Name	Design and Analysis of Algorithms Lab	**Course Code**	**25CAP-612**

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
0	0	4	0	2.00	P

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Practical Examination (PRAC)	Practical (PRAC)

Mission of the Department	M1 To provide innovative learning centric facilities and quality-oriented teaching learning process for solving computational problems. M2 To provide a frame work through Project Based Learning to support society and industry in promoting a multidisciplinary activity. M3To develop crystal clear evaluation system and experiential learning mechanism aligned with futuristic technologies and industry. M4 To provide doorway for promoting research, innovation and entrepreneurship skills in collaboration with industry and academia. M5 To undertake societal activities for upliftment of rural/deprived sections of society
Vision of the Department	To be a Centre of Excellence for nurturing computer professionals with strong application expertise through experiential learning and research for matching the requirements of industry and society instilling in them the spirit of innovation and entrepreneurship.

Program Educational Objectives(PEOs)	
PEO1	Demonstrate analytical and design skills including the ability to generate creative solutions and foster team-oriented professionalism through effective communication in their careers.
PEO2	Expertise in successful careers based on their understanding of formal and practical methods of application development using the concept of computer programming languages and design principles in accordance to industry 4.0
PEO3	Exhibit the growth of the nation and society by implementing and acquiring knowledge of upliftment of health, safety and other societal issues.
PEO4	Implement their exhibiting critical thinking and problem- solving skills in professional practices or tackle social, technical and business challenges.

Program Specific OutComes(PSOs)	
PSO1	Analyze their abilities in systematic planning, developing, testing and executing complex computing applications in field of Social Media and Analytics, Web Application Development and Data Interpretations.
PSO2	Apprise in-depth expertise and sustainable learning that contributes to multi-disciplinary creativity, permutation, modernization and study to address global interest.

Program OutComes(POs)	
PO1	Apply mathematics and computing fundamental and domain concepts to find out the solution of defined problems and requirements.
PO2	Use fundamental principle of Mathematics and Computing to identify, formulate research literature for solving complex problems, reaching appropriate solutions.
PO3	Understand to design, analyse and develop solutions and evaluate system components or processes to meet specific need for local, regional and global public health, societal, cultural, and environmental systems.
PO4	Use expertise research-based knowledge and methods including skills for analysis and development of information to reach valid conclusions.
PO5	Adapt, apply appropriate modern computing tools and techniques to solve computing activities keeping in view the limitations
PO6	Exhibiting ethics for regulations, responsibilities and norms in professional computing practices.

PO7	Enlighten knowledge to enhance understanding and building research, strategies in independent learning for continual development as computer applications professional.
PO8	Establishing strategies in developing and implementing ideas in multi- disciplinary environments using computing and management skills as a member or leader in a team.
PO9	Contribute to progressive community and society in comprehending computing activities by writing effective reports, designing documentation, making effective presentation, and understand instructions
PO10	Apply mathematics and computing knowledge to access and solve issues relating to health, safety, societal, environmental, legal, and cultural issues within local, regional and global context.
PO11	Gain confidence for self and continuous learning to improve knowledge and competence as a member or leader of a team.
PO12	Learn to innovate, design and develop solutions for solving real life business problems and addressing business development issues with a passion for quality competency and holistic approach.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Introduction to the Design and Analysis of Algorithms	Anany Levitin	2nd Edition	Pearson	2009
2	Computer Algorithms	Sartaj Sahni and Ellis Horowitz	1st Edition	Computer Science Press	1997

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L.	3rd Edition	PHI (Prentice Hall of India) / MIT Press	2009
2	Design and Analysis of Algorithms	S. Sridhar	First Edition	Oxford University Press (Higher Education	2007

Course OutCome	
SrNo	OutCome
CO1	Understand the basics of different data structures to manage the data
CO2	Analyze the asymptotic performance of algorithms through algorithmic complexity of simple, non-recursive programs.
CO3	Understand the fundamentals of data structures
CO4	Apply and analyze important algorithmic design paradigms and their applications.
CO5	Implement the major graph algorithms to model engineering problems.

Lecture Plan Preview-Practical

Unit No	ExperimentNo	Experiment Name	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer(s)
1	1	Sort a given set of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO1
1	2	Sort a given set of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO1
1	3	Sort a given set of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO1

1	4	Sort a given set of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO1
1	5	Using Open, implement a parallelized Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO2
1	6	Using Open, implement a parallelized Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO2

1	7	Using Open, implement a parallelized Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO2
1	8	Using Open, implement a parallelized Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO2
2	9	Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal’s algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO3
2	10	Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal’s algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO3
2	11	Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal’s algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO3
2	12	Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal’s algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO4

2	13	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO4
2	14	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO4
2	15	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO4
2	16	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO4
3	17	a. Implement 0/1 Knapsack problem using dynamic programming. b. Compute the transitive closure of a given directed graph using Warshall's algorithm	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	18	a. Implement 0/1 Knapsack problem using dynamic programming. b. Compute the transitive closure of a given directed graph using Warshall's algorithm	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	19	a. Implement 0/1 Knapsack problem using dynamic programming. b. Compute the transitive closure of a given directed graph using Warshall's algorithm	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	20	a. Implement 0/1 Knapsack problem using dynamic programming. b. Compute the transitive closure of a given directed graph using Warshall's algorithm	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5

3	21	Find a subset of a given set S={sl,s2, ,sn} of n positive integers whose sum is equal to a given positive integer d. For example, if S={1,2,5,6,8}and d = 9 there are two solutions{1,2,6}and {1,8}.A suitable message is to be displayed if the given problem instance doesn't have a solution.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO4
3	22	Find a subset of a given set S={sl,s2, ,sn} of n positive integers whose sum is equal to a given positive integer d. For example, if S={1,2,5,6,8}and d = 9 there are two solutions{1,2,6}and {1,8}.A suitable message is to be displayed if the given problem instance doesn't have a solution.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	23	Find a subset of a given set S={sl,s2, ,sn} of n positive integers whose sum is equal to a given positive integer d. For example, if S={1,2,5,6,8}and d = 9 there are two solutions{1,2,6}and {1,8}.A suitable message is to be displayed if the given problem instance doesn't have a solution.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	24	Find a subset of a given set S={sl,s2, ,sn} of n positive integers whose sum is equal to a given positive integer d. For example, if S={1,2,5,6,8}and d = 9 there are two solutions{1,2,6}and {1,8}.A suitable message is to be displayed if the given problem instance doesn't have a solution.	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	25	Project Title: Air Traffic Control	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5

3	26	Project Title: Air Traffic Control	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	27	Project Title: Air Traffic Control	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	28	Project Title: Air Traffic Control	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	29	Project Title: Air Traffic Control	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5
3	30	Project Title: Air Traffic Control	,T- Computer Algorithms,T-Introduction to the Design and,R- Introduction to Algorithms,R-Design and Analysis of Algorit	Simulation Practical	CO5

Assessment Model			
Sr No	Assessment Name	Exam Name	Max Marks
1	20PRAC01	External Viva / Voce	40
2	20PRAC01	Experiment-1	30
3	20PRAC01	Experiment-2	30
4	20PRAC01	Experiment-3	30
5	20PRAC01	Experiment-4	30
6	20PRAC01	Experiment-5	30
7	20PRAC01	Experiment-6	30
8	20PRAC01	Experiment-7	30
9	20PRAC01	Experiment-8	30
10	20PRAC01	Experiment-9	30
11	20PRAC01	Experiment-10	30
12	20PRAC01	Mid-Term Test	15

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	NA	NA	1	1	NA	NA	NA	NA	NA	NA	NA	NA	1
CO2	3	NA	3	2	2	NA	NA	NA	NA	NA	NA	NA	1	1
CO3	3	NA	3	2	3	NA	NA	NA	NA	NA	NA	NA	1	1
CO4	2	3	NA	1	2	NA	NA	NA	NA	NA	NA	NA	1	3
CO5	2	NA	2	2	3	NA	NA	NA	NA	NA	NA	NA	3	2
Target	2.6	3	2.67	1.6	2.2	NA	NA	NA	NA	NA	NA	NA	1.5	1.6